

Workload SLO Recommendations

Reading-speed derivation (chat-tier ITL)

The 50 ms ITL target for human-streamed output comes from the following:

- Average adult reading speed is 220–350 WPM (scholarwithin.com).
- Take 400 WPM as a fast reader (high end of normal).
- Apply a 5× skim multiplier (the model should out-pace the reader, not match them) → 2000 WPM.
- Convert: 2000 words/min × 1.3 tokens/word ≈ 2600 tokens/min ≈ 23 ms/token.
- Round to **50 ms** as the p99 ceiling.

This is consistent with MLPerf v5.1's Interactive tier of 30–40 ms p99 TPOT.

Reference Anchors

Anchor	Value	Source
Average adult reading	220–350 WPM	scholarwithin.com
"Faster than reading"	100 ms TPOT ≈ 450 WPM	Databricks LLM Inference blog
MLPerf v5.1 Interactive (Llama 8B/70B)	TTFT ≤ 500 ms p99, TPOT ≤ 30–40 ms p99	MLCommons v5.0 announcement
MLPerf v5.0/5.1 Server (Llama 8B/70B)	TTFT ≤ 2 s p99, TPOT ≤ 100–200 ms p99	MLCommons v5.0 announcement
MLPerf v5.0 Llama 405B Server	TTFT ≤ 6 s p99, TPOT ≤ 175 ms p99	MLCommons v5.0 announcement
Chatbot "feels responsive"	TTFT < 500 ms; premium < 200 ms	Databricks LLM Inference blog , AWS Bedrock latency-optimized
Batch / once-daily report	E2E ~30 s acceptable	Databricks LLM Inference blog
SLOs-Serve Tight / Loose	50 ms / 100 ms TPOT	arXiv 2504.08784

Per-Workload SLOs

interactive-chat

Metric	p99	Source
TTFT	500 ms	MLPerf v5.1 Interactive p99 ≤ 500 ms; matches Databricks and Bedrock "feels responsive"
ITL	50 ms	Reading-speed derivation; MLPerf Interactive

code-generation

Metric	p99	Source
TTFT	30 s	MLPerf Server baseline (2 s) extended ~15× because the lognormal ISL distribution has p99 near the 990 k cap; prefill is compute-bound and dominates first-token latency at this scale
ITL	100 ms	MLPerf Server tier (100 ms TPOT). Reading-speed does not apply: code is reviewed in blocks, not streamed prose, so the chat-tier 50 ms is over-spec'd

deep-research

Metric	p99	Source
TTFT	10 s	MLPerf Server baseline (2 s) extended ~5× because exponential ISL is capped at

Metric	p99	Source
		150 k as the loop accumulates context
ITL	100 ms	MLPerf Server tier (100 ms TPOT). Reading-speed does not apply for an agentic loop where most turns are machine-consumed

reasoning

Metric	p99	Source
TTFT	2 s	MLPerf Server p99 ≤ 2 s; ISL mean 1 k makes prefill fast
ITL	100 ms	MLPerf Server tier (100 ms TPOT). Thinking phase is not user-visible (only matters for total wall-clock); final-answer reading-speed is dominated by total OSL, not per-token rate

batch-summarization-rag

Metric	p99	Source
TTFT	15 s	MLPerf Server tier (2 s) extended $\sim 7\times$ because ISL is bimodal up to 80 k (full-document reads dominate prefill) and the user expects a "summarizing..." wait
ITL	100 ms	Output is structured/JSON, parsed by code rather than read live, so MLPerf Server tier (100–200 ms) applies instead of reading-speed
E2E	6 min	TTFT (15 s) + p99 OSL (2 k) $\times$ ITL (100 ms) ≈ 215 s minimum; doubled to 6 min for queueing and prefill jitter on loaded systems. Databricks batch anchor (~ 30 s once-daily report) is the lower bound for short-doc cases, but the bimodal long-doc tail dominates p99

batch-synthetic-data-generation

Metric	Target	Source
Goodput	tokens/s/GPU (or tokens/s/\$)	MLPerf Offline treats this scenario as throughput-only; latency targets here actively hurt by preventing high-batch regimes
E2E p99	30 min	Runaway backstop only. OSL up to 8 k $\times$ 100 ms ITL ≈ 13 min worst-case generation; 30 min provides $\sim 2\times$ margin for queueing and prefill on long inputs without biting under normal load

Final Summary

Workload	TTFT p99	ITL p99	E2E p99
interactive-chat	500 ms	50 ms	—
code-generation	30 s	100 ms	—
deep-research	10 s	100 ms	—
reasoning	2 s	100 ms	—
batch-summarization-rag	15 s	100 ms	6 min
batch-synthetic-data	—	—	30 min